

DECISION SUPPORT SYSTEM FOR RANKING BEST STUDENTS USING WEIGHTED PRODUCT AT SMK BINA INDUSTRI CIKARANG

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Abstract

This study is driven by the manual and conventional process of selecting the best students at SMK Bina Industri Cikarang through deliberation forums, which leads to high subjectivity, potential calculation errors in fragmented data, and time inefficiency. The objective of this research is to design and develop a web-based Decision Support System (DSS) that provides objective, accurate, and transparent student ranking results. The research methodology employed is Rapid Application Development (RAD), encompassing stages of Requirements Planning, User Design with iterative prototyping cycles, Construction, and Cutover. The system implements the Weighted Product (WP) method as the core algorithm to evaluate student criteria systematically. The expected results of this study include a digital platform that integrates assessment data and automates the ranking process using a modern tech stack consisting of Next.js, TypeScript, and PostgreSQL. In conclusion, the application of the RAD methodology and Weighted Product algorithm within this system is expected to minimize reliance on manual decisions and significantly improve administrative efficiency in the school environment.

Keywords: Decision Support System, Weighted Product, Rapid Application Development, Next.js, SMK Bina Industri Cikarang.

1. Introduction

Vocational education has a strategic role in preparing students with practical skills that correspond to industrial needs. In this context, recognizing outstanding students is important because it can encourage academic motivation, discipline, and competitiveness. However, selecting the best student is not a simple administrative activity because the assessment must combine several criteria with different levels of importance. Student achievement is not only reflected through academic scores, but also through attitude, organizational participation, and performance in internship or field-work-practice activities.

Information technology can reduce subjectivity and improve administrative efficiency through the implementation of a Decision Support System (DSS). A DSS allows data to be processed systematically, supports decision makers with structured information, and minimizes errors that commonly occur in manual calculation. In educational institutions, a web-based DSS is particularly relevant because assessment data can be stored centrally, calculated automatically, and accessed more efficiently by authorized users.

At SMK Bina Industri Cikarang, the selection of the best student is currently conducted through teacher deliberation involving teachers and school representatives. Although deliberation remains valuable as a managerial activity, the absence of standardized quantitative calculation may produce different interpretations among assessors. The process also requires considerable time because assessment data must be collected, compared, discussed, and summarized manually before a final recommendation is produced.

Another problem is the fragmentation of assessment documents. Academic scores, attitude evaluations, organizational activity data, and internship scores are managed separately by different school units. This condition makes data consolidation slower and increases the possibility of input errors or repeated verification. As a result, the ranking process may be delayed and the basis of the final decision may be difficult to explain transparently to students and stakeholders.

The Weighted Product (WP) method is suitable for this problem because it evaluates alternatives by multiplying attribute ratings after each rating is raised to the power of its normalized criterion

weight. This mechanism enables each criterion to contribute proportionally to the final preference score. Prior research has shown that WP can be applied to selection problems involving multiple criteria and can produce objective ranking recommendations when criterion weights are defined appropriately [1], [2].

Based on these problems, this research aims to design a web-based DSS for best student selection at SMK Bina Industri Cikarang using the Weighted Product method. The proposed system is not intended to replace the authority of teachers or school management. Instead, it functions as a decision-support instrument that provides transparent, measurable, and efficient ranking recommendations.

1.1. Problem Statement

1. The selection process is still dominated by teacher deliberation and therefore may contain subjective judgment.
2. Assessment data are stored in separate documents, making consolidation slow and error-prone.
3. There is no automated mechanism for calculating criterion weights and producing student rankings.

1.2. Research Objectives

1. To design a web-based DSS that centralizes assessment criteria data for best student selection at SMK Bina Industri Cikarang.
2. To implement the Weighted Product method for calculating normalized weights, preference values, and student rankings.
3. To prepare a system development and testing framework that can support the next implementation stage of the thesis.

1.3. Significance and Scope of the Study

This study is significant because it provides a structured solution for improving transparency and administrative efficiency in student achievement selection. The scope is limited to SMK Bina Industri Cikarang, four assessment criteria determined by the school, and a web-based implementation concept using TypeScript, Next.js, and PostgreSQL. The output expected from the system is a ranked recommendation list from the highest to the lowest preference value, while the final decision remains under school authority.

2. Literature Review

2.1. Decision Support System

A Decision Support System is a computer-based information system used to process data and provide interactive information for decision making [3]. A DSS is useful for semi-structured decision problems because it combines data and analytical models to assist decision makers [4]. In this research, the DSS concept is applied to best student selection, where several assessment variables must be calculated objectively before a recommendation can be produced.

2.2. Weighted Product Method

The Weighted Product method is a Multiple Attribute Decision Making approach that evaluates alternatives based on independent attributes. The method connects attribute ratings through multiplication, where each rating is raised to the power of its corresponding normalized weight [5]. This characteristic makes WP appropriate for ranking alternatives because the influence of each criterion is represented directly in the final preference value. Studies on student achievement and other selection cases indicate that WP can improve objectivity and transparency in ranking processes [1], [6].

2.3. Rapid Application Development

Rapid Application Development (RAD) is a software development model that emphasizes short and iterative development cycles. The model uses prototyping and continuous user feedback to ensure that the system corresponds to user requirements [7]. RAD is suitable for this study because the

school can provide feedback during user interface design, requirement validation, and prototype refinement before the system is fully implemented [8].

2.4. Web-Based Technology: TypeScript, Next.js, and PostgreSQL

TypeScript is a superset of JavaScript that provides static typing and supports more structured application development [9], [10]. Next.js is a React-based framework for building modern full-stack web applications with server-side rendering and performance optimization features [11], [12]. PostgreSQL is an open-source relational database management system capable of handling structured data and supporting data integrity through relational database features [13], [14]. These technologies are selected to support a web-based DSS with centralized data processing and reliable calculation logic.

2.5. Unified Modeling Language

Unified Modeling Language (UML) is a standardized modeling language used to visualize, specify, construct, and document software systems [15]. UML diagrams, such as use case diagrams, activity diagrams, sequence diagrams, and class diagrams, can assist system designers in describing user interactions, system behavior, and database structure [16]. In this research, UML is planned to support the design of the proposed DSS before the implementation stage.

2.6. Related Studies and Research Gap

Several previous studies have applied the Weighted Product method to decision-support problems. However, the research object, system architecture, and implementation technology differ from this study. Table 1 summarizes the related studies and the gap addressed by this research [1], [17], [18], [19], [20].

Table 2.1. Related Studies and Research Gap

No	Researcher / Year	Research Focus	Method	Research Gap Addressed
1	Mardian et al. (2023)	Outstanding student selection	Weighted Product	Different object and more modern full-stack implementation using Next.js.
2	Supardi and Sono (2023)	Best employee selection	Weighted Product	Different domain and web-based PostgreSQL architecture.
3	Natanael and Kusumaningsih (2021)	Best organization member selection	Weighted Product	Different subject and use of TypeScript-based implementation.
4	Magdalena and Prihatini (2021)	Employee bonus decision	Weighted Product	Different output focus and educational ranking context.
5	Hermansyah et al. (2023)	Strategic housing location selection	Weighted Product	Different decision domain and school-based DSS problem.

Source: Research Results (2026)

The gap addressed in this study is the development of a web-based DSS for selecting the best student at SMK Bina Industri Cikarang using the Weighted Product method. Unlike previous works with different objects or older implementation contexts, this study emphasizes a school-specific ranking problem, centralized assessment data, and a modern web stack.

3. Research Methodology

3.1. Research Design

This research uses an applied research approach because the main objective is to design a technological solution for a practical decision-making problem at SMK Bina Industri Cikarang. The system development model used in this study is Rapid Application Development. RAD was selected because it supports rapid prototyping, user involvement, and iterative refinement, which are necessary for aligning the DSS with school requirements.

3.2. Research Object and Participants

The research object is the best-student selection process at SMK Bina Industri Cikarang, located in West Cikarang, Bekasi Regency. Because the study is an applied system-development study rather than a statistical survey, the population and sample are not treated as quantitative sampling units. The main informants are selected purposively based on their relevance to the selection process, namely the school principal and curriculum staff. These informants provide information about the existing selection mechanism, assessment criteria, and administrative problems.

3.3. Data Collection Methods

Data are collected through three methods. First, observation is conducted to understand the existing best-student selection procedure and identify problems related to manual data management. Second, interviews are conducted with school representatives to obtain assessment criteria, criterion weights, and system requirements. Third, literature study is conducted by reviewing journals and books related to DSS, Weighted Product, RAD, web-based development, database management, and UML.

3.4. Research Instruments

The instruments used in this research include observation notes, interview guidelines, literature review records, requirement checklists, and calculation validation sheets. The validation sheets are prepared to compare manual WP calculation results with the system-generated results in the later implementation stage. These instruments support requirement identification, system design, and calculation accuracy assessment.

3.5. Criteria and Weight Scale

The criteria are determined based on interviews with the school. All criteria in this study are classified as benefit attributes because higher values indicate better student performance. Table 2 presents the criteria used in the proposed DSS. The initial criterion importance is determined using a five-level scale.

3.6. Weighted Product Data Analysis Technique

The data analysis technique uses the Weighted Product calculation procedure. The first step is normalizing the criterion weights so that the total normalized weight equals one. The formula is shown below.

$$W_j = \frac{w_j}{\sum w_j} [21]$$

In the formula, w_j represents the normalized weight of criterion j , W_j represents the initial criterion weight, and $\sum(W_j)$ represents the total weight of all criteria.

The second step is calculating the S vector for each alternative. Because all criteria are benefit attributes, the normalized weights are applied as positive exponents.

$$S_i = \prod_{j=1}^n X_{ij}^{W_j} [21]$$

Here, S_i represents the multiplicative score of alternative i , x_{ij} represents the rating value of alternative i on criterion j , and w_j represents the normalized weight of criterion j .

The final preference value is calculated through the V vector formula.

$$V_i = \frac{\prod_{j=1}^n x_{ij}^{W_j}}{\prod_{j=1}^n (x_{ij})^{W_j}} [21]$$

The students are ranked from the highest V_i value to the lowest V_i value. The highest value becomes the primary recommendation for the best student candidate. This result is used as decision-support information and not as an automatic replacement for school management judgment.

3.7. System Development Procedure

The requirements planning stage identifies functional and non-functional requirements, including login, student data management, criteria management, weight management, WP calculation, ranking display, and report output. The user design stage prepares interface designs and receives feedback

from school users. The construction stage implements the approved design into a web-based system using TypeScript, Next.js, and PostgreSQL. The cutover stage prepares final testing, user training, and deployment activities for the complete thesis implementation.

3.8. Research Framework

The research framework connects the input, process, and output of the study. The input stage consists of observation, interviews, literature review, criteria, and initial weights. The process stage consists of RAD-based development and Weighted Product calculation. The output stage is a web-based DSS prototype that generates transparent student ranking recommendations.

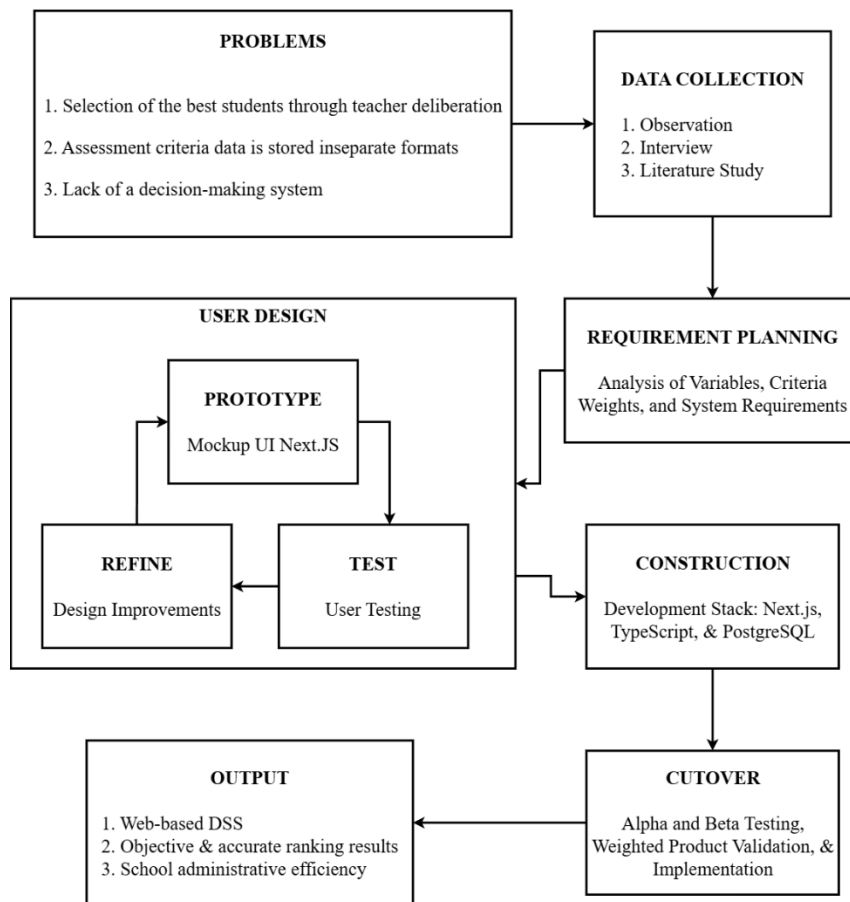


Figure 3.1. Research Framework

Source: Research results (2026)

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